

EXPLORATORY DATA ANALYSIS REPORT

E-Commerce Orders Dataset

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1. Problem Statement

This report presents an Exploratory Data Analysis (EDA) of an e-commerce orders dataset provided by DecodeLabs as part of the Data Analytics Internship, Project 2. The main objective of this analysis is to understand the structure of the data and uncover patterns, trends, and anomalies that can support better business decisions.

The core questions this analysis seeks to answer is: What are the key patterns, trends, and anomalies present in the e-commerce order data that can help improve business performance?

To answer this, the analysis covers product performance, customer payment behaviour, order status distribution, referral source effectiveness, revenue trends over time, and outlier detection in order values.

2. Dataset Overview

The dataset contains 1,200 order records with 14 columns. The data spans from January 2023 to June 2025 and covers various aspects of the ordering process including product details, customer information, payment methods, and order status.

Table 1: Dataset Overview

Column Name	Data Type	Description
OrderID	Text	Unique identifier for each order
Date	Date	Date the order was placed
CustomerID	Text	Unique identifier for each customer
Product	Text	Name of the product ordered
Quantity	Integer	Number of units ordered
UnitPrice	Decimal	Price per unit of the product
ShippingAddress	Text	Delivery address of the customer
PaymentMethod	Text	Payment method used by the customer
OrderStatus	Text	Current status of the order
TrackingNumber	Text	Shipment tracking number
ItemsInCart	Integer	Total number of items in the cart
CouponCode	Text	Discount coupon applied

ReferralSource	Text	Channel through which the customer found the store
TotalPrice	Decimal	Total monetary value of the order

3. Methodology

The analysis was carried out using Python in a Jupyter Notebook environment. The libraries used include pandas for data manipulation, matplotlib and seaborn for data visualization, and numpy for numerical computations. The following steps were performed during the analysis:

- **Data Loading:** The Excel file was loaded into a pandas DataFrame and a copy was created to preserve the original data.
- **Data Inspection:** The shape, column names, and data types were checked using `df.info()` and `df.describe()`.
- **Missing Value Treatment:** 309 missing values were found in the CouponCode column and were filled with the label No Coupon.
- **Descriptive Statistics:** Mean, median, minimum, maximum, and standard deviation were calculated for numeric columns.
- **Categorical Analysis:** Value counts were computed for Product, PaymentMethod, OrderStatus, ReferralSource, and CouponCode columns.
- **Data Visualization:** Bar charts, histograms, line charts, a boxplot, and a heatmap were created to support the analysis.
- **Outlier Detection:** The IQR method was applied to the TotalPrice column to identify unusually high or low order values.
- **Trend Analysis:** Monthly order trends were extracted from the Date column to observe patterns over time.
- **Correlation Analysis:** Pearson correlation was computed between numeric columns to identify relationships.

4. Descriptive Statistics

4.1 Summary of Numeric Columns

The table below shows the key statistical measures for the numeric columns in the dataset.

Table 2: Summary of Numeric Columns

Metric	Quantity	UnitPrice	ItemsInCart	TotalPrice
Count	1200	1200	1200	1200
Mean	2.95	356.41	5.49	1053.97
Median	3.00	364.21	5.00	823.62
Minimum	1	11.39	1	11.39
Maximum	5	699.93	10	3456.40
Std Deviation	1.41	197.18	2.28	819.86

The mean TotalPrice of 1053.97 is noticeably higher than the median of 823.62. This difference suggests that the TotalPrice distribution is right-skewed, meaning a small number of high-value orders are pulling the average upward. In such cases, the median is a more reliable representation of the typical order value. The same pattern is slightly visible in UnitPrice as well.

5. Categorical Analysis

5.1 Orders by Product

The table below shows the number of orders placed for each product category.

Table 3: Orders by Product

Product	Order Count	Percentage
Printer	181	15.1%
Tablet	179	14.9%
Chair	178	14.8%
Laptop	173	14.4%
Desk	170	14.2%
Monitor	163	13.6%

Phone	156	13.0%
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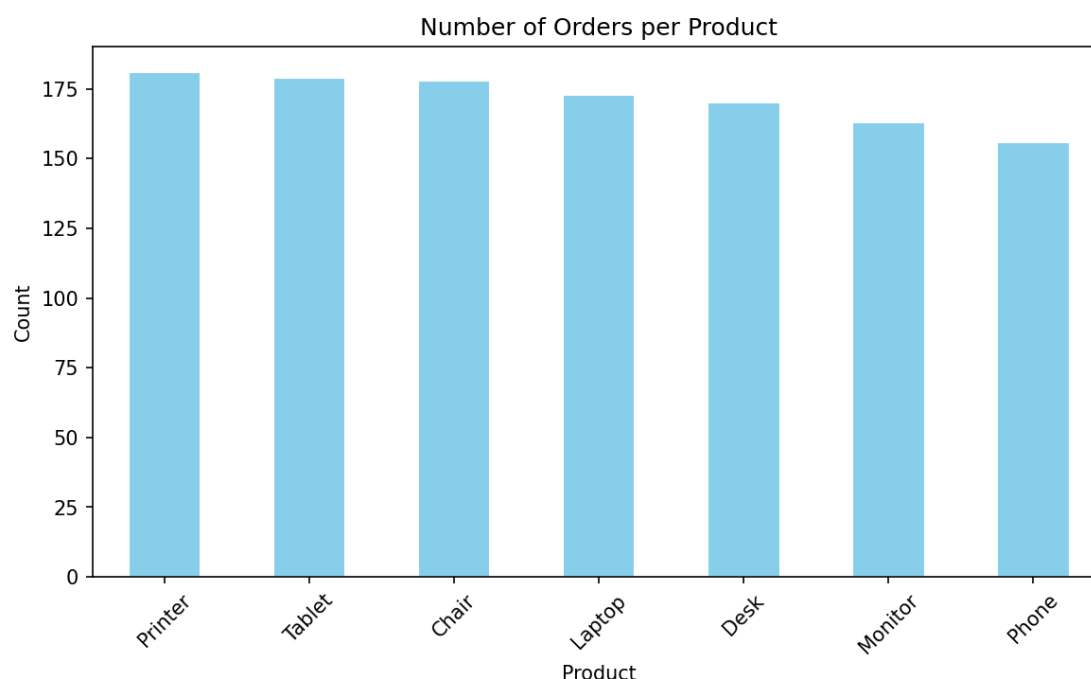


Figure 1: Bar Chart showing number of orders per product category

The distribution of orders across products is relatively balanced. Printer leads with 181 orders while Phone has the lowest count at 156. The difference between the highest and lowest selling product is only 25 orders, which indicates that no single product dominates the sales. This suggests a healthy and diversified product portfolio.

5.2 Order Status Distribution

The table below summarizes how orders are distributed across different status categories.

Table 4: Order Status Distribution

Order Status	Count	Percentage
Cancelled	250	20.8%
Returned	247	20.6%
Pending	237	19.8%
Shipped	235	19.6%
Delivered	231	19.3%

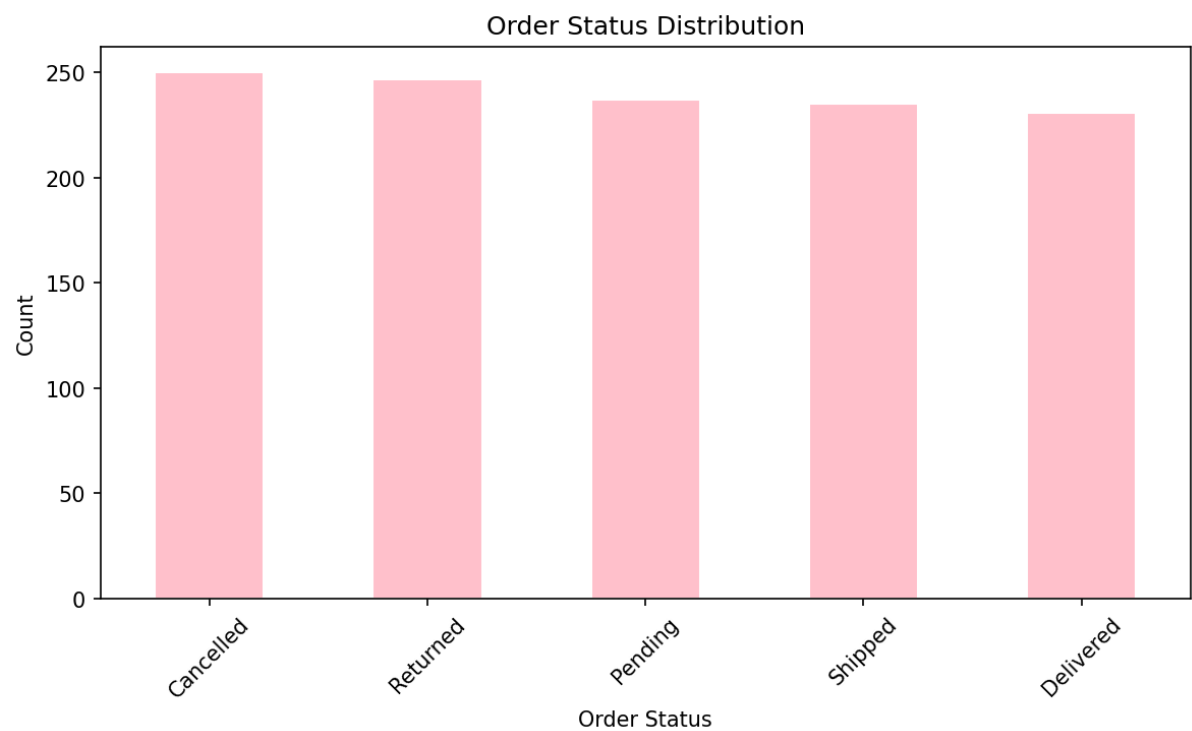


Figure 2: Bar Chart showing distribution of order statuses

This is one of the most important findings in the entire analysis. Cancelled and Returned orders together account for 41.4% of all orders. In contrast, only 19.3% of orders have been successfully Delivered. This is a significant concern for the business. A high cancellation and return rate usually point to issues such as poor product quality, misleading product descriptions, delayed delivery, or unmet customer expectations. This area requires immediate investigation and corrective action.

5.3 Order by Referral Source

The table below shows which marketing channels are bringing the most customers to the store.

Table 5: Order Status by Referral Source

Referral Source	Count	Percentage
Instagram	259	21.6%
Email	250	20.8%
Google	241	20.1%
Facebook	228	19.0%
Referral	222	18.5%



Figure 3: Bar Chart showing number of orders by referral source

Instagram is the leading customer acquisition channel, contributing 21.6% of all orders. However, all five channels are closely distributed, with the difference between the highest and lowest being just 37 orders. This shows that the business has a well-balanced and diversified marketing strategy, and no single channel is being overly relied upon.

5.4 Payment Method Distribution

The table below shows the frequency of each payment method used by customers.

Table 6: Payment Method Distribution

Payment Method	Count	Percentage
Online	258	21.5%
Cash	246	20.5%
Credit Card	234	19.5%
Debit Card	232	19.3%
Gift Card	230	19.2%

Online payment is the most used method, followed closely by Cash. All five payment methods are used in roughly equal proportions, which indicates that customers have

diverse payment preferences and the platform supports multiple payment options effectively.

5.5 Coupon Code Usage

The table below shows how coupon codes are distributed among orders.

Table 7: Coupon Code Usage

Coupon Code	Count	Percentage
FREESHIP	313	26.1%
No Coupon	309	25.75%
WINTER15	292	24.3%
SAVE10	286	23.8%

FREESHIP is the most frequently used coupon, suggesting that customers are highly motivated by free shipping offers. Around 25.75% of customers placed orders without using any coupon at all. This represents an opportunity to increase coupon adoption through better promotion and targeted discount campaigns.

6. Outlier Detection

6.1 IQR Method Applied on TotalPrice

The Interquartile Range (IQR) method was used to detect outliers in the TotalPrice column. This method is robust against extreme values and is well-suited for business data.

Table 8: IQR Method

Metric	Value
Q1 (25th Percentile)	410.52
Q3 (75th Percentile)	1578.48
IQR (Q3 - Q1)	1167.96
Lower Bound (Q1 - 1.5 x IQR)	-1341.41
Upper Bound (Q3 + 1.5 x IQR)	3330.41
Number of Outliers Detected	8

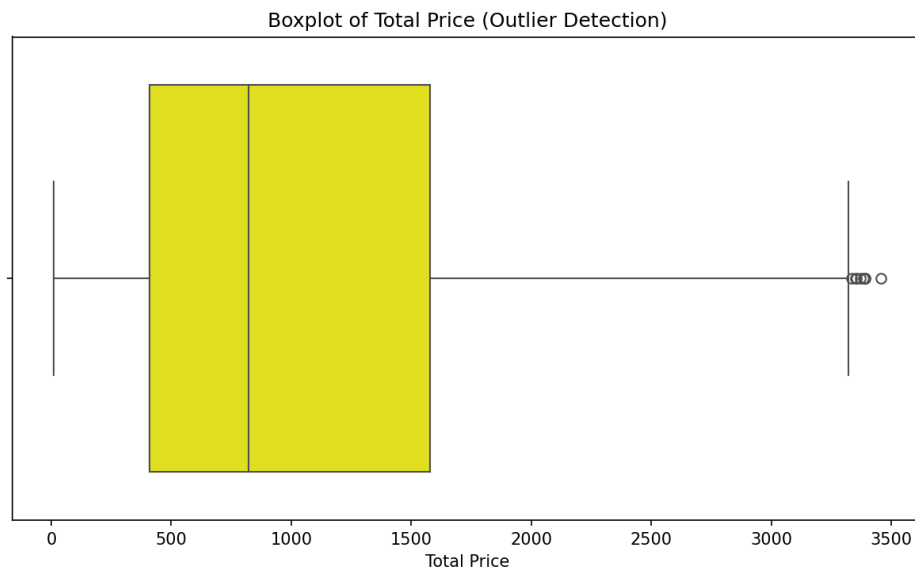


Figure 4: Boxplot of TotalPrice showing outlier distribution

A total of 8 orders were identified as outliers, all with TotalPrice values above 3330.41. Since the lower bound is negative and no order has a negative price, there are no low-end outliers in the dataset. These 8 high-value orders could represent legitimate bulk purchases from corporate or institutional customers, or they could be the result of data entry errors. Each of these orders should be reviewed individually to determine whether they are valid transactions or need to be corrected.

7. Distribution of TotalPrice

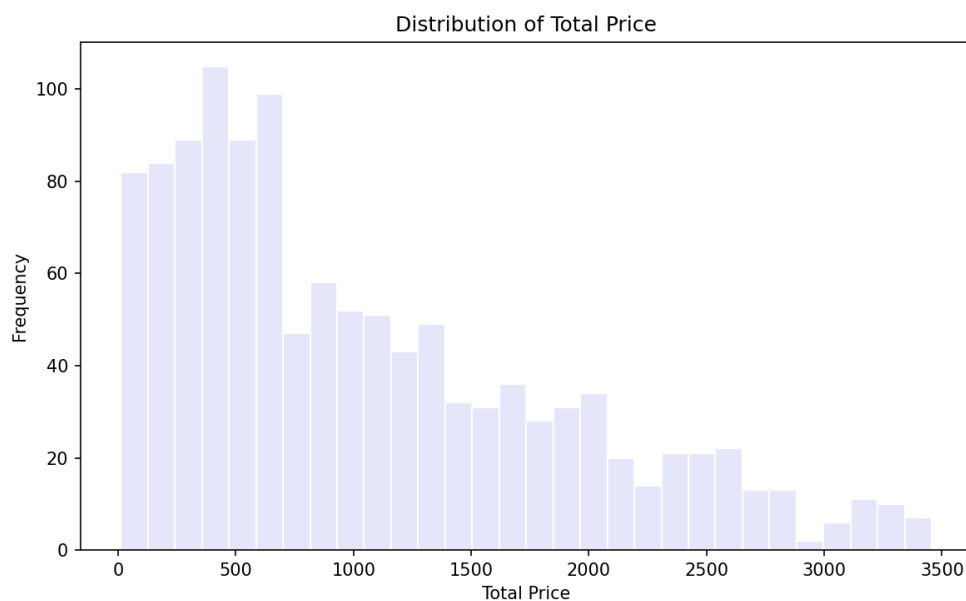


Figure 5: Histogram showing the distribution of TotalPrice

The histogram of TotalPrice confirms that the data is right-skewed. Most orders fall in the lower to mid-price range with very few high-value orders. This supports the presence of outliers and explains why the mean exceeds the median.

8. Monthly Sales Trend

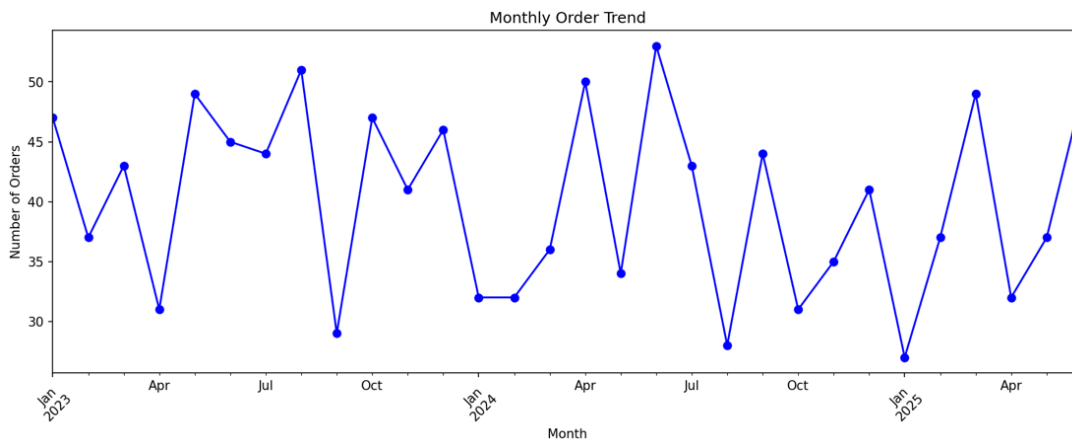


Figure 6: Line Chart showing monthly order count from January 2023 to June 2025

The monthly order trend was analyzed from January 2023 to June 2025. Order volumes fluctuate across months with no clear seasonal pattern or consistently dominant month.

9. Total Revenue by Product

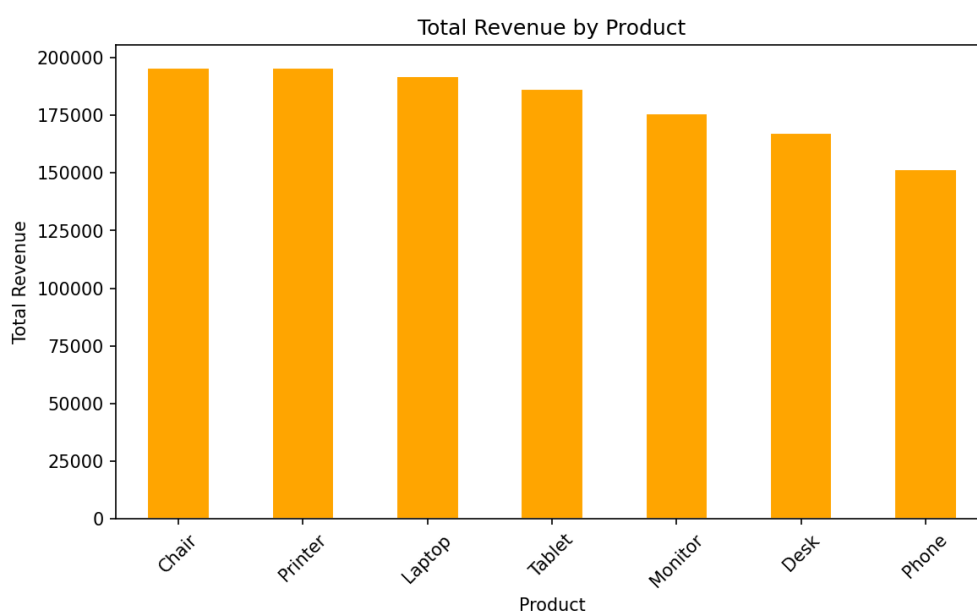


Figure 7: Bar Chart showing total revenue generated by each product

While Printer had the highest order count, the total revenue generated by each product depends on both the quantity sold and the unit price. This chart provides a clearer picture of which products are contributing more to the overall business revenue.

10. Correlation Analysis

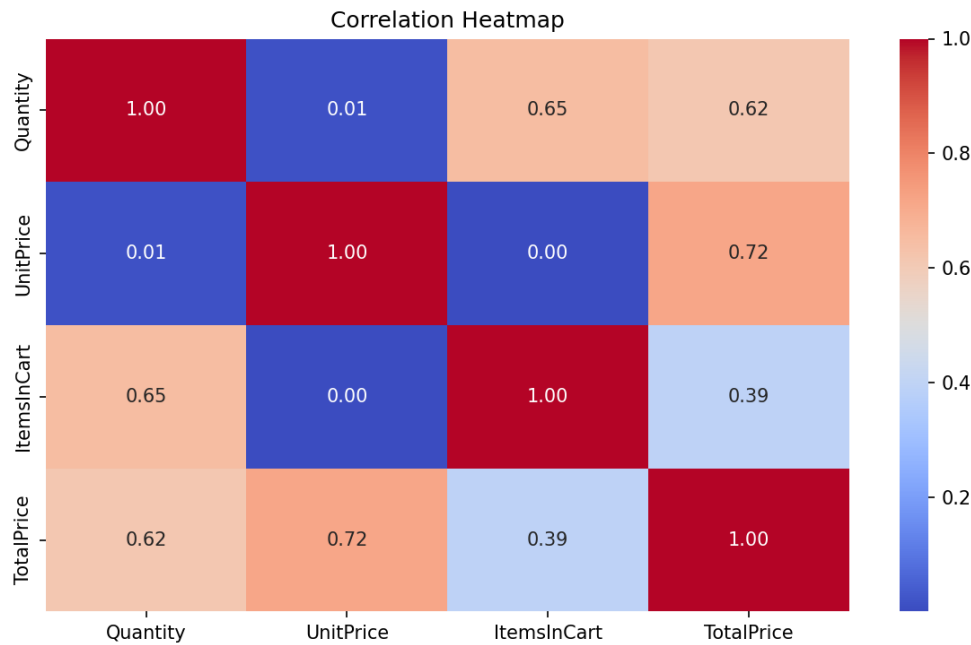


Figure 8: Heatmap showing correlation

Pearson correlation was computed between the four numeric columns in the dataset. The key observations from the heatmap are as follows. UnitPrice and TotalPrice show a positive correlation, meaning that orders with higher unit prices tend to have higher total values. Quantity also positively correlates with TotalPrice since buying more units naturally increases the total. ItemsInCart shows a weaker correlation with TotalPrice, indicating that having more items in the cart does not always result in a higher order value.

It is important to note that correlation does not imply causation. These relationships are useful clues that suggest where further investigation may be valuable, but they do not confirm a direct cause-and-effect relationship between variables.

11. Key Findings and Business Diagnosis

The table below summarizes the most important findings from the analysis along with their corresponding business implications.

Table 9: Key Findings and Business Diagnosis

Finding	Business Implication
41.4% of orders are Cancelled or Returned	Urgent review of product quality, delivery process, and customer service is needed
Only 19.3% of orders are successfully Delivered	The order fulfillment pipeline is significantly underperforming
Instagram drives the most traffic at 21.6%	Increasing marketing investment on Instagram could yield stronger returns
Median order value is 823.62, not 1053.97	Most customers spend below the average, so mid-range promotions would be more effective
8 outlier orders found above 3330.41	These orders need individual review to confirm validity or identify data errors
25.75% of orders used no coupon	Expanding coupon campaigns could attract more budget-conscious customers
Printer is the top-selling product by order count	Sufficient inventory and related accessory promotions should be prioritized for Printers
FREESHIP is the most used coupon	Free shipping is a strong motivator and should be promoted more widely

12. Recommendations

Based on the findings of this analysis, the following actions are recommended for the business:

- Investigate the root causes of the high cancellation and return rates by conducting customer feedback surveys and reviewing order fulfillment processes.
- Strengthen the Instagram marketing strategy since it is the top customer acquisition channel and consider running product-specific campaigns there.
- Expand coupon and discount campaigns to reach the 25.75% of customers who currently place orders without any coupon.

- Use the median order value of 823.62 as the benchmark for pricing strategy and promotional planning rather than the mean, which is inflated by outliers.
- Review the 8 high-value outlier orders individually to confirm whether they are valid bulk purchases or data errors that need correction.
- Analyze which specific products are associated with the highest cancellation and return rates to identify problem areas within the product catalog.
- Introduce a loyalty or rewards program to encourage repeat purchases from existing customers identified through the CustomerID column.
- Monitor monthly revenue trends in addition to order counts to get a more accurate picture of business performance over time.

13. Conclusion

This Exploratory Data Analysis of the e-commerce orders dataset successfully uncovered several meaningful insights from what initially appeared to be a straightforward table of numbers. The analysis followed a structured approach from data loading and cleaning, through statistical computation and visualization, to the synthesis of actionable business findings.

Among the most important discoveries is the alarmingly high rate of cancelled and returned orders, which together account for over 41% of all transactions. This is a critical operational issue that should be prioritized. At the same time, positive signals were also identified, including a well-balanced product mix, diversified marketing channels, and strong customer engagement through the FREESHIP coupon.

This project demonstrates that the true value of data analysis lies not in calculating statistics, but in translating those numbers into clear, actionable business language. A data analyst serves as the bridge between raw data and informed decision-making, and this analysis fulfills that role effectively.